

Final term project of SMMG 6000M – Robotics in Manufacturing

April. 13, 2026

(2nd version)

1. Project Description

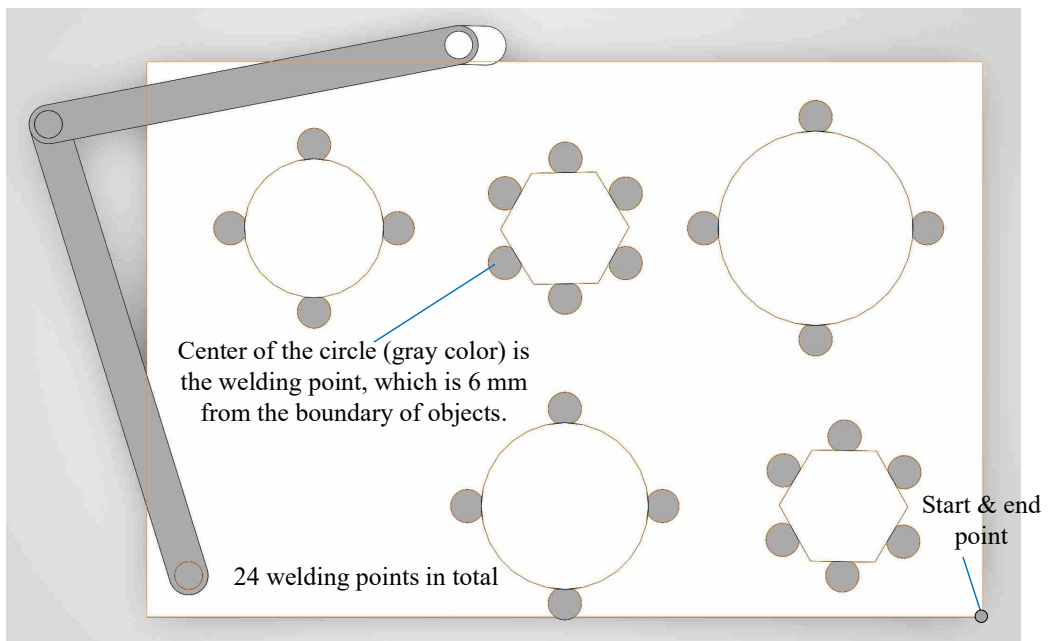
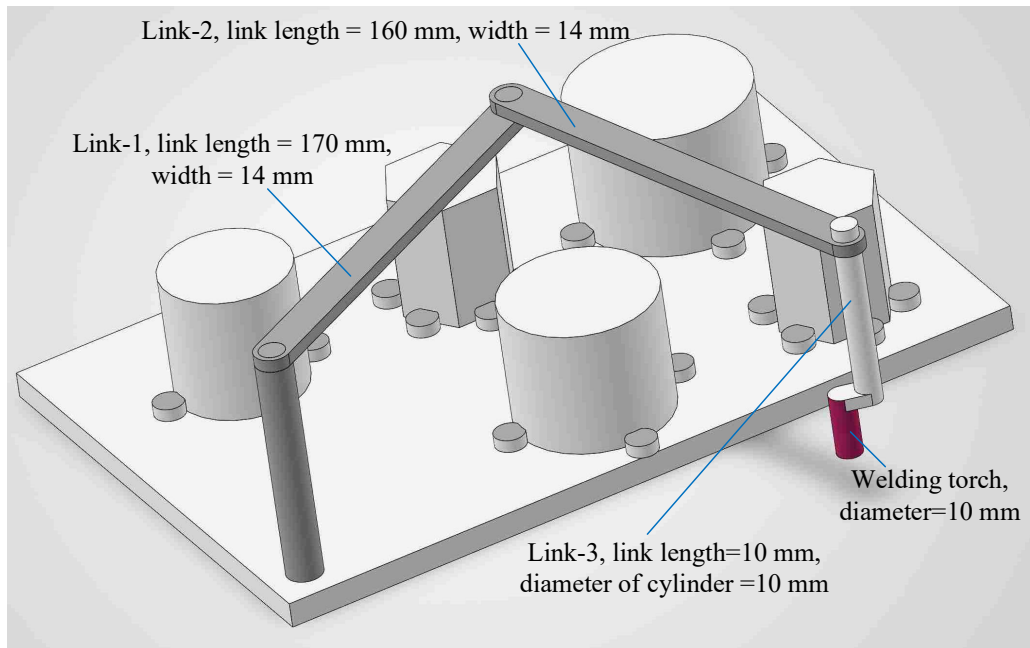
This project integrates the essential topics covered in the SMMG6000M – Robotics in Manufacturing course. Students will engage in a real-world robotic manufacturing scenario, where they are tasked with designing and implementing key technologies related to path planning, trajectory planning, and both inverse and forward kinematics of a typical industrial robot.

The project will be conducted **in teams of five students**.

2. Project Tasks

The focus of this project is a robotic welding task involving an RRR robot equipped with a welding torch as the end-effector (welding torch is simplified as a cylinder of radius = 10 mm, as shown in Fig. 1). There are in total five objects precisely positioned on a plane, and the robot must navigate to all the 24 welding points on boundaries of these objects to firmly weld them on the plane. The specific requirements for the welding task are as follows:

- (1) At each welding point, Link 3 of the RRR robot must be perpendicular to the boundary of the object.
- (2) Given the specific start and end point of the RRR robot, the total path length should be as short as possible.
- (3) Throughout the entire process, there must be no interference between the welding torch, the objects, and the links of the robotic arm.



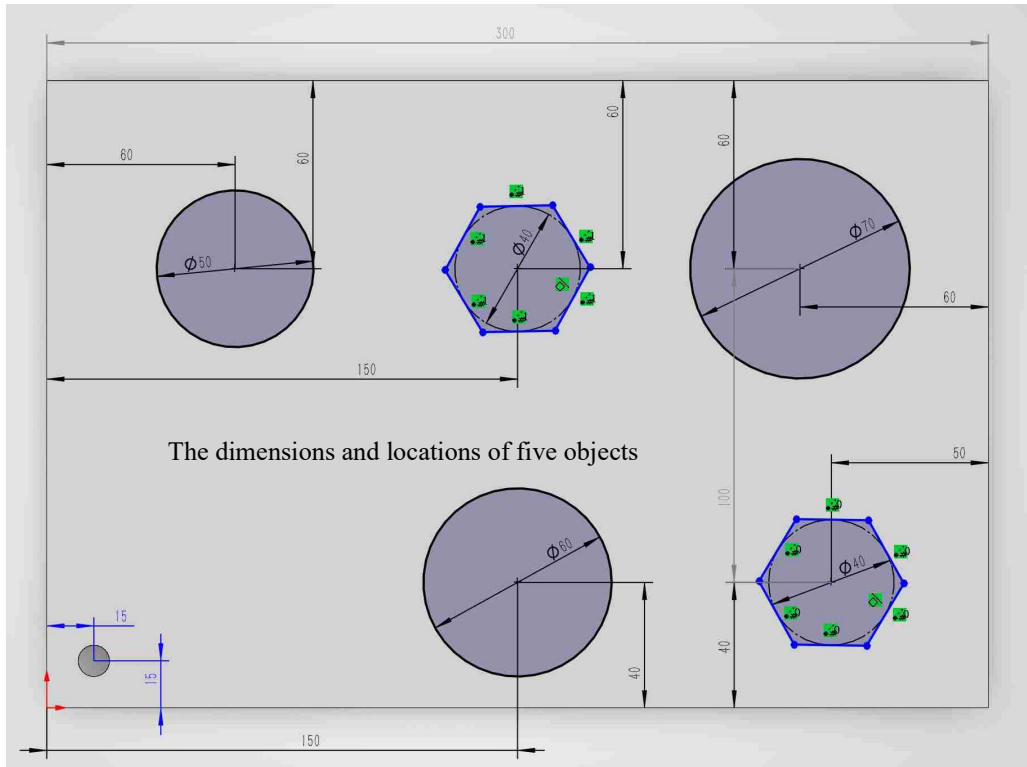


Fig. 1 The robotic welding task.

3. Task details

Task 1: Welding Point Sequence Determination

Identify the optimal sequence of all the welding points (of all objects) to minimize the path length. (**Hint:** Hamiltonian cycle in graph theory.)

Task 2: Path Generation

Based on the welding point sequence from Task 1, plan a collision-free path between each pair of neighboring points. (**Hint:** May utilize methods such as the potential field method, RRT-based method, graph-based method, or others - choose one.)

Task 3: Trajectory Planning

Given the path generated in Task 2, develop the trajectory in both joint space and Cartesian space using cubic polynomials or a linear function blended with parabolic segments.

Task 4: Collision Checking and Avoidance

For the trajectory planned in Task 3 (in both joint space and Cartesian space), check for potential collisions between the welding torch, the objects, and the robotic arm. The clearance of collision detection is 1mm. If a collision is detected, modify the path from

Task 2 and the trajectory from Task 3 until a collision-free solution is achieved.

Task 5: Project Reporting

Each team will submit a single project report and create a PowerPoint presentation clearly delineating the contributions of each member. All team members must sign the final submitted report to confirm their contributions.

4. Deliverables

Each group is required to submit the following:

- A complete set of code for the project (C++, C, Python, Matlab, Java, etc.).
- A comprehensive project report and accompanying PowerPoint presentation.
- A group presentation scheduled for April 28 during the regular lecture time, with each group allocated 12 minutes for presentation followed by a 5-minute Q&A session.
- All deliverables must be submitted by 6 PM on April 28. Please compress all materials into a single ZIP file, naming it according to your group name (e.g., Group1-Final_Term_Project.zip), and upload it to Canvas.
- The use of AGI tools is permitted during project implementation and report drafting, but must be clearly identified in the final submission.